- Typical fuel characteristics can be obtained from the fuel supplier.

Fuel parameters outside of the Cat fuel specification limits have explainable consequences.

- Some fuel parameters that are outside of the specification limits can be compensated for (e.g. fuel can be cooled to address low viscosity; etc.).
- Some fuel parameters that are outside of specification limits may be able to be improved with the use of appropriate amounts of well proven fuel additives. Refer to this Special Publication, “Distillate Diesel Fuel” article, “Aftermarket Fuel Additives” and “Cat Diesel Fuel Conditioner” topics for guidance.

To help ensure optimum engine performance, a complete fuel analysis should be obtained before engine operation. The fuel analysis should include all of the properties that are listed in the “Caterpillar Specification for Distillate Fuel for Nonroad Diesel Engines”, Table 51 .

Note: The diesel fuel has to be bright and clear. The diesel fuel cannot have any visually apparent sediment, suspended matter, or undissolved water.

Diesel Fuels that meet the specifications in table 51 will help provide maximum engine service life and performance.

In North America, diesel fuels that are identified as meeting the latest version of “ASTM D975” Grades No. 1-D or No. 2-D (all listed sulfur levels) generally meet the table 51 requirements.

In Europe, diesel fuels that are identified as meeting the latest version of “European Standard EN590” generally meet the table 51 requirements.

Table 51 is for diesel fuels that are distilled from conventional sources (crude oil, shale oil, oil sands, etc.). Diesel fuels from other sources could exhibit detrimental properties that are not defined or controlled by this specification.

NOTICE

Ultra Low Sulfur Diesel (ULSD) fuel 0.0015 percent (≤15 ppm (mg/kg)) sulfur is required by regulation for use in engines certified to nonroad Tier 4 standards (U.S. EPA Tier 4 certified) and that are equipped with exhaust aftertreatment systems.

European ULSD 0.0010 percent (≤10ppm (mg/kg)) sulfur fuel is required by regulation for use in engines certified to European nonroad Stage IIIB and newer standards and are equipped with exhaust aftertreatment systems.

Certain governments/localities and/or applications MAY require the use of ULSD fuel. Consult federal, state, and local authorities for guidance on fuel requirements for your area.

Typical aftertreatment systems include Diesel Particulate Filters (DPF), Diesel Oxidation Catalysts (DOC), Selective Catalytic Reduction (SCR) and/or Lean NOx Traps (LNT). Other systems may apply.

Low sulfur diesel (LSD) fuel 0.05 percent (≤500 ppm (mg/kg) sulfur) is strongly recommended for use in engines that are pre-Tier 4 models, while diesel fuel with > 0.05 percent (500 ppm (mg/kg)) sulfur is acceptable for use in areas of the world where allowed by law. Pre-Tier 4 engines that are equipped with a Diesel Oxidation Catalyst (DOC) require the use of LSD fuel or ULSD fuel.

ULSD fuel or sulfur-free diesel fuel are applicable for use in all engines regardless of the engine Tier or Stage.

Use appropriate lubricating oils that are compatible with the engine certification and aftertreatment system and with the fuel sulfur levels. Refer to the “Diesel Fuel Sulfur Impacts” article of this “Fuels Specifications” section and to the “Lubricants Specifications” section of this Special Publication.

Recommendations for Europe Stage V Certified Nonroad Engines:

All the fuel recommendations and requirements for US EPA Tier 4 Certified Nonroad Engines” section above are applicable to the Europe Stage V type-approved Nonroad Engines. Additionally, for the correct operation of the engine in order to maintain the gaseous and particulate pollutant emissions of the engine within the limits of the type-approval, unless specified otherwise in the engine-specific Operation and Maintenance Manual, EU Stage V regulations REQUIRE the diesel fuels (also called non-road gas oil) used in engines operated within the European Union (EU) to have the characteristics below:

- The sulfur content should be ≤ 10 mg/kg (20 mg/kg) at point of final distribution
- The Cetane number should be ≥ 45

PROOF

- The biodiesel (also called Fatty Acid Methyl Ester (FAME)) content should be ≤ 7 % volume/volume

Refer to your engine Operation and Maintenance Manual for the allowed biodiesel blend in your engine model. Some Cat engines that are certified per EU Stage V can use up to B20 biodiesel blends.

Follow all the local regulations and fluids requirements in your area. Refer to your engine-specific Operation and Maintenance Manual, and refer to your aftertreatment device documentation, if available, for additional guidance.

Engine operating conditions play a key role in determining the effect that fuel sulfur will have on engine deposits and on engine wear.

Note: The removal of sulfur and other compounds in Ultra Low Sulfur Diesel (ULSD) fuel decreases the conductivity of ULSD and increases the ability of the fuel to store static charge. Refineries may have treated the fuel with a static dissipating additive. However, there are many factors that can reduce the effectiveness of the additive over time. Static charges can build up in ULSD fuel while the fuel is flowing through fuel delivery systems. Static electricity discharge when combustible vapors are present could result in a fire or explosion. Therefore, ensuring that the entire system used to refuel your machine (fuel supply tank, transfer pump, transfer hose, nozzle, and others) is properly grounded and bonded is important. Consult with your fuel or fuel system supplier to ensure that the delivery system is in compliance with fueling standards for proper grounding and bonding practices.

NOTICE
Do not add new engine oil, waste engine oil or any oil product to the fuel unless the engine is designed and certified to burn diesel engine oil (for example Caterpillar ORS designed for large engines). Caterpillar experience has shown that adding oil products to Tier 4 engine fuels (U. S. EPA Tier 4 certified), to EURO Stage IIB and IV certified engine fuels, or to the fuels of engines equipped with exhaust aftertreatment devices, will generally cause the need for more frequent ash service intervals and/or cause loss of performance.

Adding oil products to the fuel may raise the sulfur level of the fuel and may cause fouling of the fuel system and loss of performance.

ULSD and any other fuel used in Cat engines have to be properly formulated and addetized by the fuel supplier and have to meet Special Publication, “Caterpillar Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines”. Fuels that are defined as “ASTM D975” Grade No. 1-D S15 or “ASTM D975” Grade No. 2-D S15 generally meet Cat requirements for ULSD.

Refer to this Special Publication, “Characteristics of Diesel Fuel” article for additional pertinent information concerning fuel lubricity, fuel oxidative stability, fuel sulfur, and aftertreatment devices. Also refer to the latest version of “ASTM D975” , the latest version of “EN 590”, the specific engine Operation and Maintenance Manual, and to aftertreatment device documentation for guidance.

Note: Caterpillar strongly recommends the filtration of distillate fuel and/or biodiesel/biodiesel blends through a fuel filter with a rating of four microns(c) absolute or less. This filtration should be on the device that dispenses the fuel to the fuel tank for the engine, and also on the device that dispenses fuel from the bulk storage tank. Series filtration is recommended. Caterpillar recommends that the fuel dispensed into the machine tank meets “ISO 18/16/13” cleanliness level.

Note: The owner and the operator of the engine has the responsibility of using the correct fuel that is recommended by the manufacturer and allowed by the U.S. EPA and, as appropriate, other regulatory agencies.

NOTICE
Operating with fuels that do not meet Cat recommendations can cause the following effects: starting difficulty, reduced fuel filter service life, poor combustion, deposits in the fuel injectors, reduced service life of the fuel system, deposits in the combustion chamber and reduced service life of the engine.

NOTICE
The footnotes are a key part of the “Caterpillar Specification for Distillate Diesel Fuel” Table. Read ALL of the footnotes.

For additional guidance related to many of the fuel characteristics that are listed, refer to “Cat Specification for Distillate Fuel for Off-Highway Diesel Engines”, table 51 .

The values of the fuel viscosity given in table 51 are the values as the fuel is delivered to the fuel injection pumps. For ease of comparison, fuels should also meet the minimum and maximum viscosity requirements at 40° C (104° F) that are stated by the use of either the “ASTM D445” test method or the “ISO 3104” test method. If a fuel with a low viscosity is used, cooling of the fuel may be required to maintain 1.4 cSt or greater viscosity at the fuel injection pump. Fuels with a high viscosity might require fuel heaters to lower the viscosity to either 4.5 cSt or less for rotary fuel injection pumps or 20 cSt viscosity or less for all other fuel injection pumps.

UNCONTROLLED

Table 51

Cat Specification for Distillate Fuel for Nonroad Diesel Engines			
Specifications	**Requirements**	**ASTM Test**	**ISO Test**
Aromatics	35% volume, maximum	“D1319”	“ISO 5186”
Ash	0.01% maximum (weight)	“D482”	“ISO 6245”
Density at 15° C (59° F)(1)(2)	800 kg/m^3 minimum 860 kg/m^3 maximum	“D4052”, “D287”	“ISO 3675”, “ISO 12185”
Cetane Number	40 minimum (DI engines)	“D613”	“ISO 5165”
	35 minimum (PC engines)		
Cetane Index	40, minimum	“D976”	“ISO 4264”
Flash Point	legal limit	“D93”	“ISO 2719”
Carbon Residue on 10% distillation residue - Ramsbottom, % mass	0.30% mass, maximum	“D524”	“ISO 10370”
Oxidation Stability	25 g/m^3, maximum	“D2274”	“ISO 12205”
Thermal Stability	Minimum of 80% reflectance after aging for 180 minutes at 150° C (302° F)	“D6468”, “D3241”	No equivalent test
Copper Strip Corrosion (Control temperature 50° C (122° F) minimum)	No. 3 maximum	“D130”	“ISO 2160”
Distillation, vol recovered	10%, Record	“D86”	“ISO 3405”
	90% at 360° C (680° F) maximum (3)		
	90% at 350° C (662° F) maximum (3)		
Lubricity (HFRR Wear Scar)	0.52 mm (0.0205 inch) maximum at 60° C (140° F)	“D6079”/“D7688”	“ISO 12156–1.3”
Pour Point	6°C (10°F) minimum below ambient temperature	“D97”	
Cloud Point	The cloud point must not exceed the lowest expected ambient temperature.	“D2500”	“ISO 3015”
Low Temperature Flow Test/Cold Filter Plugging Point Test (LTFT/CFPP)	Must not exceed the lowest expected ambient temperature.	“D4539”/“D6371”	No equivalent test
Sulfur by weight	(4)	“D5453”, “D2622”, “D129” (based on the sulfer level)	“ISO 20846”, “ISO 20884”
KInematic Viscosity at 40° C (104° F) for No. 1 diesel	1.3 cSt minimum and 2.4 cSt maximum	“D445”	“ISO 3104”
KInematic Viscosity at 40° C (104° F) for No. 2 diesel	1.9 cSt minimum and 4.5 cSt maximum	“D445”	“ISO 3104”
Contaminants			
Solids	10 mg/l	“D6217”	“ISO 12662”
Sediment	0.05% maximum (weight)	“D473”	No Equivalent Test

(continued)

PROOF

(Table 51, contd)

Cat Specification for Distillate Fuel for Nonroad Diesel Engines			
Specifications	**Requirements**	**ASTM Test**	**ISO Test**
Water/Sediment	0.05% maximum	“D2709”	“ISO 3734”
Water	0.02% maximum	“D1744”	“ISO 12937”
Cleanliness	(5)	“D7619”	“ISO 4406”
Appearance	Clear and Bright	“D4176”	No equivalent test

(1) The equivalent API gravity of 875.7 kg/m^3 is 30 and for 801.3 kg/m^3 is 45 (per “ASTM D287” test method temperature of 15.56° C (60° F)).
(2) The density range allowed included # 1 and # 2 diesel fuel grades. Fuel density varies depending on sulfur levels, where high sulfur fuels have higher densities. Some unblended (neat) alternative fuels have lower densities than diesel fuel. This density is acceptable if the other properties of the alternative fuel fall within this specification.
(3) Distillation of 90% at 350° C (662° F) maximum is recommended for Tier 4 engines and preferred for all engines. Distillation of 90% at 350° C (662° F) is equivalent to 95% at 360° C (680° F). Distillation of 90% at 360° C (680° F) maximum is 360° C (680° F). Distillation of 90% at 360° C (680° F) maximum is acceptable for Pre-Tier 4 engines.
(4) Follow the federal, state, local, and other governing authorities for guidance concerning the fuel requirements in your area. Follow the engine Operation and Maintenance Manual and the details provided in this Fuel section. ULSD 0.0015% (<15 ppm S) is required by law for Tier 4 engines and engines with aftertreatment devices. ULSD and LSD 0.05% (≤500 ppm S) are strongly recommended for pre-Tier 4 engines. Diesel fuel with >0.05% (>500 ppm) sulfur is acceptable for use where allowed by law. Consult your Cat dealer for guidance when sulfur levels are above 0.1% (1000 ppm). Certain Cat fuel systems and engine components can operate on fuel with a maximum sulfur content of 3%. Refer to the specific engine Operation and Maintenance Manual and consult your Cat dealer
(5) Recommended cleanliness level for fuel as dispensed into machine or engine fuel tank is “ISO 18/16/13” or cleaner per “ISO 4406” or “ASTM D7619”. Refer to the “Recommendations for Cleanliness of Fuels” in this chapter

There are many other diesel fuel specifications that are published by governments and by technological societies. Usually, those specifications do not review all the requirements that are addressed in the “Caterpillar Specification for Distillate Fuel for Off-Highway Diesel Engines”, Table 51 . To help ensure optimum engine performance, a complete fuel analysis should be obtained before engine operation. The fuel analysis should include all the properties that are listed in the “Cat Specification for Distillate Fuel for Off-Highway Diesel Engines”, Table 51 .

NOTICE
In order to meet expected fuel system component life, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with common-rail fuel systems. Also, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with unit injected fuel systems. For all other Cat diesel engines (mostly older engines with pump, line and nozzle type fuel systems), the use of 4 micron(c) absolute or less secondary fuel filtration is strongly recommended. Note that all current Cat diesel engines are factory equipped with Cat Advanced Efficiency 4 micron(c) absolute fuel filters.

To obtain additional information on Cat designed and produced filtration products, refer to this Special Publication, “Reference Material” article, “Filters” and “Miscellaneous” topics, and then contact your Cat dealer for assistance with filtration recommendations for your Cat machine.

⚠ WARNING

Mixing alcohol or gasoline with diesel fuel can produce an explosive mixture in the engine crankcase or fuel tank.

Personal injury and damage to the engine may result. Caterpillar recommends against this practice.

Caterpillar S·O·S Services Fuel Analysis

Caterpillar has developed a maintenance management tool that evaluates fuels. The Cat tool for fuel analysis is called S·O·S Fuel Analysis and is part of the S·O·S Services Program.

Testing the diesel fuel that goes into your engine is an important tool in your equipment management toolkit. Diesel fuel testing can help identify production limiting issues such as rapid fuel filter plugging, hard starting, white smoke, deposits, accelerated wear, and low power. Diesel fuel testing can also provide extra benefits including helping to identify fuel saving steps, environmental regulation compliance in countries with higher fuel regulations, minimizing Diesel Particulate Filter (DPF) regeneration and maximizing the life of the DPF filter and Diesel Oxidation Catalyst. Some facilities with standby generators may have requirements that fuel is tested regularly. Operations without requirements will benefit from knowing that the fuel in the standby generators is going to provide the expected performance when needed.

S·O·S Services Fuel Analyses

The Cat S·O·S Fuel Analysis program provides testing of the fuel for the properties listed below. The actual analyses provided may vary depending on your requirements and reasons for testing. Consult your local Cat Dealer for complete information and assistance about the S·O·S Fuel Analysis program.

- Biodiesel content
- Sulfur content
- Water contamination
- Particle cleanliness level
- Microbial growth
- Identification of elements that can increase deposit formation
- Identification of fuel conditions that can indicate contamination or adulteration
- Identification of fuel conditions that can indicate increased abrasive wear, adhesive wear, or wear in the combustion chamber
- Identification of fuel characteristics that can indicate low power
- Indication of fuel to perform in cold weather
- Identification of fuel conditions that can increase filter plugging
- Indication of fuel condition during storage
- Indication of ability of fuel to perform at startup

The results are reported and appropriate recommendations are provided.

A properly administered S·O·S Services Program can reduce repair costs and lessen the impact of down time. S·O·S Fuel Analysis is a key component of this program and can ensure that your fuel is stored in a clean environment, meets government requirements, and can meet the expected guidelines for performance in your engine. Consult your Cat Dealer to determine your fuel testing needs and establish a regular testing interval based on those needs. Consult the “Cat Fuel Specification” and the “Contamination Control” sections of this Special Publication for related details on fuel recommendations including cleanliness.

Obtaining S·O·S Fuel Samples

Fuel sampling methods depend on the type of fuel tank to be sampled. Storage tanks may have an automatic sampling valve at different levels. Storage tanks without an automatic sampling valve require a tank sampling device (commonly known as a “Bacon Bomb” or “Sample Thief”). Refer to “Fuel Sampling Guide”PEDJ0129, for more information and instructions on proper sampling techniques. Fuel analysis sampling kits can be obtained from your local Cat Dealer. Size of fuel sample needed may be dependent upon the list of tests required.

Heavy Fuel Oil, Residual Fuel, Blended Fuel

NOTICE
Heavy Fuel Oil (HFO), Residual fuel, or Blended fuel must **NOT** be used in Caterpillar diesel engines (except in 3600 Series HFO engines). Blended fuel is residual fuel that has been diluted with a lighter fuel (cutter stock) so that they will flow. Blended fuels are also referred to as heavy fuel oils. Severe component wear and component failures will result if HFO type fuels are used in engines that are configured to use distillate fuel.

Fuels For Cold-Weather Applications

In extreme cold ambient conditions, you may choose to use the distillate fuels that are specified in table 52 . However, the fuel that is selected must meet the requirements that are specified in the “Cat Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines”, Table 51 . These fuels are intended to be used in operating temperatures that are down to −54 °C (−65 °F).

Note: The fuels that are listed in table 52 may have higher sulfur levels than the 15 ppm maximum sulfur allowed for ULSD. The sulfur levels for these fuels may exceed 50 ppm maximum sulfur allowed in “EN590:2004”. These fuels may not be acceptable for use in areas that restrict maximum fuel sulfur levels to 15 ppm maximum or to 50 ppm maximum.

The jet fuels described in Table 52 are of lower viscosity than #2 diesel. To meet the viscosity requirements given in Table 51 , cooling of the fuel may be required to maintain 1.4 cSt or greater viscosity at the fuel injection pump. Ensure that the lubricity of these fuels is per the requirements given in Table 51 . Consult the supplier for the recommended additives to maintain the proper fuel lubricity.

PROOF

The fuel specifications listed in this table allow and/or recommend the use of fuel additives that have not been tested by Cat for use in Cat fuel systems. The use of these specifications allowed and/or recommended fuel additives are at the risk of the user.

Jet A is the standard fuel used by U.S. commercial airlines when operating within the U.S. Jet A-1 is the standard fuel used by commercial airlines worldwide. Per “ASTM D1655, Table 1 (Detailed Requirements of Aviation Turbine Fuels)”, Jet A and Jet A-1 have identical requirements except for freezing point. Jet A has a freeze point requirement of −40 °C (−40 °F) versus the Jet A-1 has a freeze point requirement of −47 °C (−52.6 °F), but the fuel purchaser and the fuel supplier may agree on other freezing points.

Table 52

Alternative Distillate Fuels - Cold-Weather Applications	
Specification	**Grade**
“MIL-DTL-5624U”	JP-5
“MIL-DTL-83133F”	JP-8
“ASTM D1655-08a”	Jet A, Jet A-1

These fuels are lighter than the No. 2 grades of fuel. The cetane number of the fuels in table 52 must be at least 40. If the viscosity is below 1.4 cSt at 40 °C (104 °F), use the fuel only in temperatures below 0 °C (32 °F). Do not use any fuels with a viscosity of less than 1.2 cSt at 40 °C (104 °F).

Note: Fuel cooling may be required to maintain the minimum viscosity of 1.4 cSt at the fuel injection pump.

Note: These fuels may not prove acceptable for all applications.

Aftermarket Fuel Additives

There are many different types of fuel additives that are available to use. Caterpillar does not generally recommend the use of fuel additives.

In special circumstances, Caterpillar recognizes the need for fuel additives. Fuel additives need to be used with caution. The additive may not be compatible with the fuel. Some additives may precipitate. This action causes deposits in the fuel system. The deposits may cause seizure. Some additives may plug fuel filters. Some additives may be corrosive, and some additives may be harmful to the elastomers in the fuel system. Some additives may damage emission control systems. Some additives may raise fuel sulfur levels above the maximum allowed by the United States (U.S.) Environmental Protection Agency (EPA) and/or, as appropriate, other regulatory agencies. Consult your fuel supplier for those circumstances when fuel additives are required. Your fuel supplier can make recommendations for additives to use and for the proper level of treatment.

Note: Metallic fuel additives can cause fuel system/injector fouling and after treatment device fouling. Caterpillar discourages the use of metallic fuel additives in most applications. Metallic fuel additives should only be used in applications where their use is specifically recommended by Caterpillar.

Note: Diesel fuel additives/conditioners may not improve markedly poor diesel fuel properties enough to make them acceptable for use.

Note: For best results, your fuel supplier should treat the fuel when additives are needed.

Diesel Fuel Conditioner

Fuels that are per specifications detailed in this publication should not require the use of additives. In case a fuel conditioner is needed to improve certain fuel properties, consult with your fuel supplier or with a reputable provider. Refer to your Cat dealer and to Cat special publication PELJ2095 for more details.

When needed, high performance, multipurpose diesel fuel conditioners are designed to improve:

- Fuel economy (through fuel system cleanup)
- Lubricity
- Oxidative stability
- Detergency/dispersancy
- Moisture dispersancy
- Corrosion protection
- Cetane (typically 2-3 cetane numbers)

Cat Diesel Fuel System Cleaner

Note: Cat Diesel Fuel System Cleaner, part number 343-6210, is the only fuel system cleaner available to the end user that is tested and approved by Caterpillar for use in Cat diesel engines.

Cat Diesel Fuel System Cleaner is a proven high-performance detergent product specifically designed for cleaning deposits that form in the fuel system. Deposits in the fuel system reduce system performance and can increase fuel consumption. Cat Diesel Fuel System Cleaner addresses the deposits formed due to the use of degraded diesel fuel, poor quality diesel fuel, and diesel fuel containing high quantities of high molecular weight compounds. Cat Diesel Fuel System Cleaner addresses deposits formed due to the use of biodiesel, biodiesel blends, and biodiesel that does not meet the appropriate quality specifications. Continued use of Cat Diesel Fuel System Cleaner is proven to inhibit the growth of new deposits.

PROOF

Cat Diesel Fuel System Cleaner can be added directly to diesel fuel, biodiesel, or biodiesel blends. Cat Diesel Fuel System Cleaner is a United States Environmental Protection Agency registered fuel additive that can be used with Ultra Low Sulfur Diesel Fuel. In addition this cleaner is appropriate for use with other ultra low, low, and higher sulfur diesel fuels around the world.

Cat Diesel Fuel System Cleaner is a proven high-performance cleaner that is designed to perform the following:

- Clean performance-robbing fuel system deposits
- Restore fuel economy losses resulting from injector deposits
- Restore power losses resulting from injector deposits
- Eliminate visible black exhaust smoke resulting from injector deposits
- Prevent the formation of new fuel-related deposits

For engines experiencing problems such as power loss, increased fuel consumption, or black smoke due to the presence of fuel-related deposits in fuel injectors, a high-strength cleaning cycle is recommended. Add one 0.946L (32 oz.) bottle of Cat Diesel Fuel System Cleaner per 250 L (65 gal) of fuel, which corresponds to a treat rate of 0.4% by volume. Prior to refueling, pour Cat Diesel Fuel System Cleaner directly into the fuel tank, then refill with fuel. The refilling process should give satisfactory mixing of the cleaner. The cleaner will begin to be effective immediately. Testing has shown most deposits are cleaned and related issues are resolved after 30 hours of operating the engine on fuel with the cleaner. For maximum results, continue to use at this treat rate for up to 80 hours.

To prevent the return of fuel-related deposits, Cat Diesel Fuel System Cleaner, add the cleaner to the fuel as previously described, but at a 0.2% treat rate. In this case, one 0.946L (32 oz.) bottle will treat 500 L (130 gallons) of fuel. Cat Diesel Fuel System Cleaner can be used on an on-going basis with no adverse impact on engine or fuel system durability.

NOTICE
Use of Cat Diesel Fuel System Cleaner or Cat Diesel Fuel System Conditioner does not lessen the responsibility of the engine owner and/or responsibility of the fuel supplier to follow all industry standard maintenance practices for fuel storage and for fuel handling. Refer to the “General Fuel Information” article in this Special Publication for additional information. Additionally, use of Cat Diesel Fuel System Cleaner or Cat Diesel Fuel System Conditioner does NOT lessen the responsibility of the owner of the engine to use appropriate diesel fuel. Refer to the “Fuel Specifications” section in this Special Publication (Maintenance Section) for guidance.

Caterpillar strongly recommends that Cat Diesel Fuel System Cleaner be used with biodiesel and biodiesel blends. Cat Diesel Fuel System Cleaner is suitable for use with biodiesel/biodiesel blends that meet Cat biodiesel recommendations and requirements. Note that not all fuel cleaners are suitable for use with biodiesel/biodiesel blends. Read and follow all applicable label usage instructions. Also, refer to this Special Publication, “Distillate Diesel Fuel”, article and also refer to the “Biodiesel” article, which includes Cat biodiesel recommendations and requirements.

When used as directed, Cat Diesel Fuel System Cleaner has proven to be compatible with non-road Tier 4 U.S. EPA certified engines that are equipped with aftertreatment devices.

Note: When used as directed, Cat Diesel Fuel System Cleaner will not raise fuel sulfur levels measurably in the final fuel/additive blend. Follow all applicable national, regional, and local laws, mandates, and regulations concerning the use of diesel fuel conditioners/additives.

NOTICE
When used as directed Cat Diesel Fuel System Cleaner will not raise fuel sulfur levels measurably in the final fuel/additive blend. But, in the U.S., aftermarket fuel additives (retail consumer level versus bulk fuel additives used at the fuel supplier/distributor level) with more than 15 ppm sulfur are NOT allowed to be used in applications where ULSD usage is mandated (15 ppm or less fuel sulfur). Note that Cat Diesel Fuel System Cleaner contains less than 15 ppm of sulfur and is acceptable for use with ULSD fuel.

Renewable and Alternative Fuels

Renewable fuels are derived from renewable resources such as planted crops and crop residues (referred to as biomass), waste, algae, cellulosic material, yard and food waste, etc. Renewable fuels reduce the carbon footprint of the fuels compared to fossil fuels on a Life Cycle Analysis basis. Caterpillar, through sustainability initiatives, supports the development and use of renewable fuels.

PROOF

Renewable fuels (other than biodiesel) and alternative fuels (such as but not limited to Gas-to-Liquid fuel) are typically >99% hydrocarbons (composed of carbon and hydrogen). An exception is biodiesel, which is an oxygenated renewable fuel. Biodiesel is discussed in a separate article in this Fuel section. Significant research is on going to develop renewable fuels and produce the fuels economically.

Caterpillar is not in a position to test all varieties of renewable and alternative fuels that are advertised in the market place. If a renewable or alternative fuel fulfills the performance requirements described in Cat Fuel Specification, the latest version of “ASTM D975”, the latest version of “EN 590”, or the latest version of the paraffinic fuel specification “CEN TS 15940” (which defines quality requirements for Gas to Liquids (GTL), Biomass to Liquids (BTL) and hydrotreated vegetable oil (HVO)), then this fuel or a blend of this fuel (blended with appropriate diesel fuel) can be used as a direct replacement of petroleum diesel in all Cat diesel engines.

Consult with the fuel supplier and with your Cat dealer to ensure that the cold-weather performance of the fuel is appropriate to the expected ambient temperatures at the operation sites and to ensure elastomer compatibility. Certain elastomers used in older engines (such as engines manufactured up to the early 1990s) may not be compatible with the new alternative fuels.

Caterpillar is following the development of renewable and alternative fuels and the respective fuel specifications to ensure successful application of these fuels in the engines. Information and guidelines will be published as the production of these fuels becomes established.

Pyrolysis Fuels

A particular family of renewable/alternative fuels, known as pyrolysis fuels, are typically NOT suitable for use in in modern diesel engines. Pyrolysis fuels can be obtained from various resources including wood, used tires, plastic, etc. Pyrolysis fuels in their raw form do not meet all the requirements in Table 51 , “ASTM D975” and/or “EN 590”specifications. These fuels have to be upgraded in order to produce a hydrocarbon product that meets all requirements defined in these specifications. Upgrading can include fractionation to remove volatiles, hydro-desulfurization, hydrotreating etc.

Where the pyrolysis fuel is obtained from wood, our experience has been that the liquid obtained contains high oxygen content (>10%), has high acidity (pH~1) and does not meet distillation, lubricity, and cetane requirements. Use of this fuel is likely to lead to severe impairment and wear in the fuel system. Suitable upgrading would likely involve, at a minimum, hydro-deoxygenation, fractionation and other hydrotreating.

Where the pyrolysis fuel is obtained from used tires, our experience has been that the distillation and sulfur requirements are not met. Use of this fuel is likely to lead to impaired function of the fuel and aftertreatment systems. Suitable upgrading would likely involve, at a minimum, fractionation and desulfurization.

Where the pyrolysis fuel is obtained from waste plastic, our experience has been that the distillation, lubricity, and cetane requirements are not met. Use of this fuel is likely to lead to impaired function of the fuel system. Suitable upgrading would likely involve, at a minimum, fractionation and other hydrotreating.

Upgraded pyrolysis fuels that meet Cat Diesel Fuel specification in Table 51 , “ASTM D975” and/or “EN 590” specifications can be considered for use in Cat diesel engines. Refer to this special publication for guidelines and requirements for fuels acceptable in Cat engines. Refer to your Cat dealer for questions.

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Biodiesel

SMCS Code: 1280

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

Biodiesel is a renewable fuel that can be made from vegetable oils, animal fat, and waste cooking oil. Soybean oil, rapeseed oil, and palm oil are some of the common vegetable oil sources. The raw oils or animal fats are chemically processed (esterified) to form a fatty acid methyl ester (referred to as FAME). The esterified product (FAME) is biodiesel fuel that can be used in compression ignition engines. Without the chemical processing referred to as esterification, the oils or fats are not suitable for use as fuel in compression ignition engines. The oil or fat must be esterified and the water and contaminants removed.

Fuel made of 100 percent FAME is referred to as B100 biodiesel or neat biodiesel.

Biodiesel can be blended with distillate diesel fuel. The blends can be used as fuel. Blends of biodiesel are represented as BXX, where XX represents the volume percent of biodiesel blended with diesel fuel.

Example:

- To blend 100 gallons of B5 biodiesel fuel: mix 5 gallons of B100 biodiesel fuel with 95 gallons of diesel fuel
- To blend 100 gallons of B20 biodiesel fuel: mix 20 gallons of B100 biodiesel fuel with 80 gallons of diesel fuel

PROOF

The quality of biodiesel fuel is critical to its successful use. Biodiesel fuel that is not per the specification given in Table 54 may result in performance issues and engine downtime.

U.S. distillate diesel fuel specification “ASTM D975” includes up to B5 (5 percent) biodiesel. Any diesel fuel in the U.S. may contain up to B5 biodiesel fuel without labeling that indicates biodiesel content in the finished fuel.

European distillate diesel fuel specification “EN 590” includes up to B5 (5 percent) biodiesel and in some regions up to B7 or B8 (7 or 8 percent by volume) biodiesel. Any diesel fuel in EU may contain up to these blend levels of biodiesel fuel without labeling that indicates biodiesel content in the finished fuel.

Certain regions or countries around the world may mandate blend levels of B20 or higher. Refer to the local regulations and mandates and to local biodiesel specifications for fuel quality.

Biodiesel fuel that is per Caterpillar and Industry recommended specifications (Refer to Table 54 in this Chapter) offers the following advantages:

- Renewable fuel, nontoxic, and biodegradable
- Reduces tailpipe particulate matter (PM), hydrocarbon (HC), and carbon monoxide (CO) emissions from most modern diesel engines
- High lubricity, hence reduces friction
- High cetane number

Biodiesel at B5 blend level has the same attributes as diesel fuel. At blend levels over B5, biodiesel has the following attributes that are different than diesel fuel and require management:

- Energy density is lower than diesel fuel. At B100, biodiesel has about 8% lower energy density than diesel fuel. At B20 or lower blend levels, the energy density difference from diesel fuel is not significant.
- Oxidation stability and storage stability are lower than diesel fuel.
- Cold temperature operability is different than diesel fuel. Cloud point pour point and CFPP are typically higher than diesel fuel.
- Materials compatibility is more restricted than diesel fuel.
- A higher tendency to dissolve and absorb water than diesel fuel.
- Metal content is higher than diesel fuel. Biodiesel can contain certain materials naturally or due to processing (phosphorus, sodium, calcium, potassium, and magnesium). The maximum levels of these materials are controlled by the appropriate specifications.
- Contaminants content due to incomplete esterification or purifying process can be present. These contaminants may include glycerides, mono and diesters, sterol glucosides, and others.
- Higher propensity for microbial growth due to the biodegradable nature of biodiesel and to the tendency of higher water absorption.

Meeting specifications as detailed in this Chapter is paramount for biodiesel fuel used in engines, to avoid performance issues and engine downtime.

Refer to guidelines given in this Chapter and to Tables 15 and 17 for the biodiesel fuel requirements, specifications and risks associated with biodiesel.

Note: The user of the engine has the responsibility of using the correct fuel that is recommended by the manufacturer. The fuel must be allowed by the U.S. EPA and other appropriate regulatory agencies.

NOTICE
In North America, the use of biodiesel from “BQ-9000” accredited producers and “BQ-9000” certified marketers is required. Refer to the “Recommendations” section for details.

NOTICE
Failures that result from the use of any fuel are not Caterpillar factory defects. Therefore, the cost of repair would NOT be covered by the Caterpillar warranty for materials and/or the warranty for workmanship.

Recommendations for the Use of Biodiesel in Caterpillar Nonroad Engines

The biodiesel blend levels acceptable for use in Cat engines in various machines, and commercial applications are given in Table 53 . Consult your Caterpillar dealer for further information on compatibility of non-metallic materials such as fuel lines, and other connections to the engine.

Biodiesel fuel must be per the quality recommendations given in NEW TABLE. Otherwise, the fuel may result in performance issues and engine downtime.

PROOF

In North America, obtain biodiesel from BQ-9000 accredited producers and BQ-9000 certified marketers. Look for the BQ-9000 biodiesel quality accreditation program certification logo that is available to distributors that meet the requirements of BQ-9000. In other areas of the world, the use of biodiesel that is BQ-9000 accredited and certified, or that is accredited and certified by a comparable biodiesel quality body to meet similar biodiesel quality control standards, is required. For more information on the BQ-9000 program, go to:

http://www.BQ-9000.org

Table 53

Recommendations for Biodiesel Fuel Application in Caterpillar Nonroad Engines (1)		
Engine Models	**Model specific**	**Biodiesel acceptable blend levels**
ACERT engines; C7 through C32; C-9 through C-18; C280 Series; CM20, CM25 and CM32; 3300 Series and 3400 Series; 3500 Series and 3600 series	Engine models with aftertreatment devices and engine models per Stage V Emissions Regulations	Up to B20
	Engine models without aftertreatment devices	Up to B20(2) (For use of higher blend levels up to B100, consult with your Cat dealer)
Cat engine model C175	Tier 4, EU Stage IIIB model, EU Stage V (Locomotive)	Up to B7
	All other C175 engines	Up to B20(2) (For use of higher blend levels up to B100, consult with your Cat dealer)
Cat engine models: 3003 through 3066	All engine models	Up to B7(3)
Cat engine models: C0.5 through C7.1(Mechanical (PLN) Fuel System)	C0.5, C0.7, C1.1, C1.5, C1.6, C2.2, C3.3, C4.4 engine models Tier 2 / EU Stage II / China NR2 or earlier Emissions Regulation, without aftertreatment devices.	Up to B7(3)
	C1.3, C1.8, C2.4, C2.6, C3.3B, C3.4 engine models without aftertreatment devices	Up to B20
	C0.5, C0.7, C1.1, C1.7, <19kW engine models, Stage V Emissions Regulations, without aftertreatment devices	Up to B20
	C0.5, C0.7, C1.1, C1.5, C1.7, C2.2, C3.3, C3.4, C3.6, C4.4, C7.1 engine models Tier 3 / EU Stage IIIA / China NR3 or later Emissions Regulation, without aftertreatment devices	Up to B20(2)
Cat engine models:C1.7 through C7.1(High Pressure Common Rail (HPCR) Fuel System)	C1.7, C2.2, C2.8, C3.4B, C3.6, C4.4, C6.6, C7.1 engine models Tier 4 / EU Stage IV / EU Stage V / China NR4 or later Emissions Regulation, with aftertreatment devices	Up to B20
	C2.8, C3.6, C4.4, C6.6, C7.1 engine models Tier 3 / EU Stage IIIA / China NR3 or later Emissions Regulation, without aftertreatment devices	Up to B20(2)
	C3.3B, C3.8 engine models with aftertreatment devices	Up to B7(3)

(continued)

PROOF

(Table 53, contd)

Recommendations for Biodiesel Fuel Application in Caterpillar Nonroad Engines (1)		
Engine Models	**Model specific**	**Biodiesel acceptable blend levels**
CAT engine models:C4.4, C6.4 and C6.6 Engine Serial Prefix 444, C4E, 666, C6E	C4.4 engines (S/N 44400001-04303)	Up to B7(3)
	C6.6 engines (S/N CE600001-14623 and S/N 66600001-09015)	Up to B7(3)
	C4.4 engines (S/N C4E05524-Up and 44404304 -Up)	Up to B20(2)
	C6.4 and certain C6.6 engines (S/N CE614624-Up and 66609016-Up)	Up to B20(2)

(1) EU Regulations require the biodiesel blends used in Stage V engines and Special-Purpose Engines (SPE) operated within the European Union (EU) to contain no more than 8% v/v FAME unless specified otherwise in the engine-specific Operation and Maintenance Manual. B8 can be used where B7 is specified.
(2) Higher blends, up to B100, where mandated can be used in these engines.
(3) Use of Biodiesel blends above 7% may cause premature wear of the fuel pump and damage to the low-pressure fuel system components as both may contain material incompatible with higher Biodiesel blends.

Table 54

Biodiesel Fuel Quality Recommendations for Caterpillar Nonroad Engines		
Fuel	**Specification(1)**	**Details**
B5(2)	Cat Distillate Diesel Fuel Specification or "ASTM D975" or "EN 590"	A blend of 5 percent by volume of B100 and 95 percent by volume of diesel fuel.
B20	"ASTM D7467" or "EN 16709" and "API" gravity 30–45	A blend of 20 percent by volume of B100 and 80 percent by volume of diesel fuel.
B30	"EN 16709" and "API" gravity 30–45	A blend of 30 percent by volume of B100 and 70 percent by volume of diesel fuel.
B100	Cat Biodiesel Fuel Specification(3) or "ASTM D6751" or "EN 14214"	Neat biodiesel fuel (100 percent) used for blending with diesel fuel to get the desired blend level.
Distillate Diesel Fuel	Cat Distillate Diesel Fuel Specification (4)	Diesel fuel used for blending with biodiesel fuel to get the desired blend level.

(1) Some regions or countries may have their own fuel specifications. Follow the local requirements and regulations and use the fuel of highest quality.
(2) The same specifications also apply to B7 and B8, which are common blend levels in some regions.
(3) Cat Biodiesel Fuel Specification (B100) is given in 55 in this Biodiesel Section of this Special Publication.
(4) Cat Distillate Diesel Fuel Specification is given in in Diesel Fuel section of this Special Publication.

Note: Do not change any engine settings when using biodiesel fuel. When the use of biodiesel fuel is planned, simply convert to this fuel. Follow the guidelines, recommendations, and quality specifications given in this Chapter to avoid any performance issues or downtime.

Two methods can be used for determining the volume percent biodiesel in a biodiesel blend:

- “ASTM D7371” - “Test Method for Determination of Biodiesel (Fatty Acid Methyl Esters) Content in Diesel Fuel Oil Using Mid Infrared Spectroscopy (FTIR-ATR-PLS Method)”
- “EN 14078” - “Liquid Petroleum Products - Determination of fatty acid methyl esters (FAME) in middle distillates -Infrared spectroscopy method”

For applications running biodiesel or biodiesel blends, if fuel treatments are needed, consult with your fuel supplier or with a reputable provider. :

Impact of Biodiesel on Engine Oil

Biodiesel fuel has higher density and lower volatility than diesel fuels. As a result, during engine operation, biodiesel fuel that dilutes the crankcase oil may not evaporate as effectively as diesel fuels. For this reason, fuel dilution of crankcase oils may be higher when biodiesel blends are used.

Also, biodiesel contains oxygen molecules. These oxygen molecules cannot be differentiated from oil oxidation when using current oil analysis techniques. As a result, biodiesel fuel dilution of the crankcase oil can appear to be higher oxidation of the oil.

PROOF

When using biodiesel fuel and higher fuel dilution and/or apparent oil oxidation are detected in the crankcase engine oil, consider all the other characteristics of the used oil If these characteristics, such as wear metals, soot, viscosity or others, are per Caterpillar guidelines and have not reached condemnation limits, then the oil drain intervals should not be impacted.

To reduce any potential impact of biodiesel fuel dilution on crankcase oil, **the use of Cat S·O·S Services oil analysis is strongly recommended when up to B20 (20 percent) and lower biodiesel blends are used, and required when using biodiesel/biodiesel blends that are B20 or above.** When requesting oil analysis, be sure to note the level of biodiesel being used (B5, B20, and so on).

Use of Biodiesel fuel in Engines with Aftertreatment Emissions Control Systems

Biodiesel fuels as defined in the current ASTM specifications, may contain phosphorous, alkali and alkaline metals (sodium, potassium, calcium, and magnesium) due to processing techniques or due to the natural contents of the biodiesel feedstock. When present in biodiesel, these metals form ash upon combustion in the engine. The ash accumulates in the aftertreatment systems such as Diesel Particulate Filters (DPF), DOC or other systems. The ash can affect the life and performance of aftertreatment emissions control devices and may cause the need for more frequent ash service intervals. For these reasons, biodiesel fuels that contain ash forming metals, even at the concentration levels defined in the current specifications, are limited to B20 blend levels in the engines with aftertreatment devices.

Local and regional regulations may also restrict the blend levels allowed in engines of certain emissions levels. **EU Stage V** regulations REQUIRE the biodiesel blends used in engines operated within the European Union (EU) to be ≤ 8 % volume/volume unless specified otherwise in the engine-specific Operation and Maintenance Manual.

Note: Note that some Cat engines that are certified per EU Stage V can use up to B20 biodiesel blends. Refer to engine or machine OMM and to Table 53 .

For Tier 4 applications in the U.S., the diesel fuel portion of the final blend must meet the requirements of S15 fuels (15 ppm sulfur) designations in the latest edition of “ASTM D975” specification. For Stage IIIB and later applications in EU, the diesel fuel portion of the final blend must meet the requirements for sulfur free (10 ppm sulfur) designation in the latest edition of “EN 590”. The final blend must have maximum of 15 ppm sulfur.

Biodiesel Fuel Stability

Biodiesel fuels key difference from diesel fuel is the ester chemical group (contains two oxygen atoms) in every biodiesel molecule. Biodiesel may also have double bonds (unsaturation) in the carbon chain that can vary based on the feedstock. Due to the ester chemical group and the unsaturation, the oxidation stability of biodiesel fuels is typically lower than that of diesel fuel. The oxidation stability of biodiesel and biodiesel blends is controlled in specifications as given in Table 54 . Biodiesel fuels that do not conform to these specifications can oxidize quickly during use due to the high temperature and pressure conditions in the engine or during storage and handling practices. Oxidized biodiesel forms acids, gums, high viscosity and deposits, which can reduce storage stability, plug filters, form deposits, particularly in the fuel injector, and reduce the performance of fuel systems. Use of **biodiesel fuels meeting or exceeding the oxidation stability limits is critical** to avoid poor performance and downtime of engines.

To avoid the issues associated with oxidized biodiesel fuel, always purchase fuels that conform to or exceed the specifications (refer to Tables 53 and 56 in this Chapter). Also, follow all the guidelines for appropriate storage and handling of this fuel such as avoiding excessive heat and sunlight during storage, exposure to oxygen (air), contact with metals such as copper, lead, tin, zinc, and others. The use of antioxidants can improve the oxidation stability of biodiesel fuel. The antioxidants are most effective when added to new fuels. Consult with your fuel supplier to ensure the quality of the fuel and refer to the details given in this chapter for guidance.

Cat Diesel Fuel System Cleaner (Part number 343 - 6210), used as needed or on an on-going basis, is most effective at cleaning and preventing the formation of fuel-related deposits.

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Refer to the “Cat Diesel Fuel System Cleaner” topics in the “Distillate Diesel Fuel” section in this Special Publication for additional information. Consult your Caterpillar dealer for availability of Cat Diesel Fuel System Cleaner. In case a fuel conditioner is needed to improve certain fuel properties, consult with your fuel supplier or with a reputable provider.

Biodiesel Fuel Storage

Storage tanks used for storing diesel fuel are appropriate for storing biodiesel fuels. Fuel storage tanks need to be cleaned thoroughly before converting to biodiesel/biodiesel blends. Conversion to biodiesel/biodiesel blends can loosen fuel system and fuel storage tank deposits. Loosened deposits result in filter plugging with the loosened deposits. For this reason, the change intervals of bulk tank continuous filtration unit, dispensing point filters, and onboard engine filters should be shortened for an extended period to allow for this cleaning effect. Once the systems are cleaned, the typical filter service intervals can be resumed.

Biodiesel fuel is hygroscopic, which implies that biodiesel tends to absorb and dissolve water at a higher concentration than diesel. All precautions and measures must be taken to ensure that storage tanks are protected from water ingress. Follow all the contamination control measures provided in the "Contamination Control" section of this Special Publication.

Biodiesel storage duration is shorter than diesel due to the chemical nature of biodiesel. Storage duration is dependent on the blend level. Blends up to B20 or B30 (where mandated) can be stored up to 8 months. Testing of biodiesel at 4 months and monthly thereafter is recommended. B100 can be stored for 4 months. Testing at 2 months and monthly thereafter is recommended. Tests should include oxidation stability, acid number, viscosity and sediments. Refer to Table 56 and related guidelines for details.

Note: If switching from biodiesel to distillate fuel, it is recommended to change all fuel filters to avoid filter plugging. Failure to do so can create low fuel pressure.

Guidelines

Information provided in Table 55 refers to biodiesel and biodiesel blends that fully comply with the appropriate specifications as described in the “Biodiesel” section of this Special Publication and to handling and maintenance procedures that follow recommended guidelines.

PROOF

Table 55

Guidelines and Potential Impacts Associated with the Use of Biodiesel and Biodiesel Blends In Cat Engines[1]				
	Impact	**B8-B20**	**B21-B30**	**B31-B100**
1	Reduction of oil change interval	No risk	Not noted in Cat engines	Medium
2	Fuel filters compatibility risk	No risk	Not noted in Cat engines	Medium risk
3	Loosening of fuel systems deposits upon conversion to biodiesel	More than ULSD	Medium	High
4	Bulk filtration of biodiesel	≤4 microns absolute, required	≤4 microns absolute required	≤4 microns absolute required
5	Energy content of biodiesel	Minor loss of 1-2 percent	Minor loss of 1-2 percent	Detectable loss of 5-8 percent
6	Elastomers compatibility	Not noted in Cat engines with required material	Not noted in Cat engines with required material	Not noted in Cat engines with required material
7	Hose compatibility	Low risk of softening	Softening may occur	Softening is likely
8	Low ambient temperature issues for both storage and operation	Gelling can start higher than 0 C	Gelling at increasing temperatures compared to diesel fuel	Gelling at increasing Temperature, -2 to 18 °C
9	Oxidation stability-Injector deposits	Not noted in Cat engines	Not noted in Cat engines	Increasing risk
10	Oxidation stability-Duration of storage [2]	8 months - start testing at 4 months[3]	8 months - start testing at 4 months[3]	4 months - start testing at 2 months[3][4]
11	Use in engines with limited operational time	Unacceptable[5]	Unacceptable[5]	Unacceptable
12	Microbial contamination and growth	Increasing. Testing required - treat at trace levels	Increasing. Testing required - treat at trace levels	High. Testing required - treat at trace levels
13	Need for water removal	Increasing	Increasing	High
14	Metal incompatibility	Incompatible with certain materials	Incompatible with certain materials	Incompatible with certain materials
15	Glycerides, total, free, mono, di (if not per spec)	Detectable impact	Detectable impact	High impact

[1] Information provided in this Table refers to biodiesel and biodiesel blends that fully comply with the appropriate specifications as described in this Chapter and per following recommended maintenance procedures.
[2] Testing of biodiesel or biodiesel blend during storage is strongly recommended. Tests should include oxidation, acid number, viscosity and sediments. Tests should be conducted periodically to ensure biodiesel is per specification. Antioxidants are allowed to improve stability. Consult with your fuel supplier for more information.
[3] Testing is recommended at the indicated duration of storage and on a monthly basis thereafter. Tests should include oxidation, acid number, viscosity and sediments.
[4] B100 should be stored at temperatures of 3 degrees C to 6 degrees C (5 degrees F to 10 degrees F) above the cloud point.
[5] If B20 or B30 are used in engines of limited operational time, it is recommended that a stabilizer additive is added at the beginning of the storage period. Follow all storage recommendations.

The following are details of the Risks and Guidelines listed in Table 55 . Refer to the paragraph reference numbers:

1. When using biodiesel fuel, dilution of oil by the fuel may increase as detailed in this Chapter. Use S·O·S Services oil analysis to monitor the condition of the engine oil. S·O·S Services oil analysis will also help determine the oil change interval that is optimum. Use of S.O.S is strongly recommended for up to B20 biodiesel blends and required for blends >B20.

2. Confirm with the filter manufacturer that the fuel filter/filters to be used are compatible with biodiesel. Fuel water separators are preferred when biodiesel is used.

Note: Cat fuel filters and Cat fuel water separators are compatible with biodiesel fuel.

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3. Conversion to biodiesel can loosen fuel tanks and fuel system deposits. During the conversion period fuel filter change intervals should be shortened to allow for this cleaning effect. Once the deposits are removed, convert back to the regular filter service intervals. Filter change interval of 50 hours or less should be used during initial conversion to B20 or higher biodiesel blends.

4. Filter biodiesel and biodiesel blends through a fuel filter with a rating of 4 microns (c) absolute or less. Filters should be on the device that dispenses the fuel from the bulk storage tank to the fuel tank for the engine. Bulk filtration with fuel water separators is recommended. Series filtration is recommended.

5. Neat biodiesel (B100) typically provides less energy per gallon compared to diesel fuels. The energy content of B100 is 5 percent to 8 percent lower than No. 2 diesel fuel. The energy content of B20 (and B30 where mandated) is 1 percent to 2 percent lower than No. 2 diesel fuel, which is not significant. Do NOT attempt to change the engine rating to compensate for the power loss. Any adjustments to the engine in service may result in violation of emissions regulations such as the U.S. EPA anti- tampering provisions. Also, if any tempering with the engine ratings occurs, problems may occur when the engine is converted back to 100 percent distillate diesel fuel.

6. Compatibility of the elastomers with biodiesel can be different than compatibility with diesel fuel. Prolonged exposure of certain elastomers, hoses, seals, and gaskets to B100 may cause some degradation and softening of these materials. The condition of gaskets, seals, and hoses should be monitored regularly. The risk of degradation increases with the increase of biodiesel blend level.

a. In general, Cat engines built early to mid-90s and beyond use Viton seals and Viton O-rings in the fuel system. Viton is compatible with biodiesel and degradation upon exposure to this fuel is not expected.

7. Nitrile hoses typically used in some fuel transfer lines are not compatible with biodiesel. Hoses exposed to biodiesel and biodiesel blends may soften and may show some beading of fluid on the outside of the hose. Monitor the condition of the hoses and confirm with the hose manufacturer that the hoses are compatible with the biodiesel blend used. If necessary, replace with hoses of compatible materials. Consult with Cat dealers for appropriate hose materials.

8. Biodiesel fuels may gel or freeze at high temperatures due to the nature of this fuel. Ensure that the biodiesel pour point is appropriate for the climate of the application. In general, the risk of low temperature gelling of biodiesel increases with the increase of blend levels and may depend on the biodiesel feedstock (soy, used cooking oil, animal fats, palm and others). If the pour point of the biodiesel is not appropriate for the climate of the application, the fuel can gel and plug filters, hoses and transfer lines. At low ambient temperatures, biodiesel fuel may need to be stored in a heated building or a heated storage tank. Consult your biodiesel supplier for assistance in the blending and attaining of the proper cloud point for the fuel. Refer to "Modifying the Cloud Point" in the "Diesel Fuel" Section of this Special Publication.

Note: The performance of cold flow improvers may be less effective in biodiesel fuel compared to diesel fuel. Consult the fuel supplier for appropriate cold flow improvers if needed.

9. Biodiesel oxidation stability is in general lower than that of diesel fuel, as detailed earlier in this Chapter. The use of biodiesel fuels that are not per specifications can accelerate fuel oxidation in the fuel system. Also, engines with an electronic fuel system operate at higher temperatures and pressures that can also accelerate fuel oxidation. Oxidized fuel can form deposits in fuel injection systems and in fuel systems in general. Always use biodiesel fuel that meets or exceeds the stability limits defined in biodiesel specifications as given in this Chapter to avoid fuel oxidation and degradation. The use of appropriate antioxidants can enhance oxidation stability of biodiesel. Refer to “Biodiesel Fuel Stability” section of this chapter for details and guidelines.

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10. Due to the chemical nature of biodiesel fuel, biodiesel can age and degrade during long-term storage. Fuel aging and fuel oxidation upon long-term storage may cause the formation of gels, acids and/or deposits. For these reasons, biodiesel should be used within a limited time from production. To ensure appropriate storage duration, testing of the stored biodiesel is recommended. Tests should include oxidation, acid number, viscosity, and sediments. Tests should be conducted at the durations given in Table 56 to ensure that biodiesel is per specification. Antioxidants are recommended to improve stability of biodiesel and increase the storage time limits. Consult with your fuel supplier or with a reputable provider to select appropriate additives.

a. B20 and B30 (where mandated) biodiesel blends can generally be stored up to 8 months. Testing of B20 and B30 blends is recommended at 4 months of storage and on a monthly basis thereafter to ensure that the fuel has not degraded.

b. B100 biodiesel can generally be stored up to 4 months. Testing of B100 is recommended at 2 months of storage and every two weeks thereafter to ensure that the fuel has not degraded. The use of appropriate additives is required if B100 is stored for more than 4 months. Consult your fuel supplier for more information.

c. B100 should be stored at temperatures of 3° C to 6° C (5° F to 10° F) above the cloud point. Other storage conditions such as avoiding direct sunlight or heat should be followed.

11. Due to limited oxidation stability and other potential issues, engines with limited operational time and seasonal use should either not use biodiesel/ biodiesel blends or, while accepting some risk, limit biodiesel to a maximum of B5. Examples of applications that should limit the use of biodiesel are the following: Standby Generator sets and certain emergency vehicles, school buses, farm machinery, etc. For more information, refer to the “Seasonal Operation” section.

a. For standby generator sets and emergency vehicles that use biodiesel, sample the fuel in the engine tank monthly. Test the fuel for acid number and oxidation stability. If the test results show that the fuel is degraded and not in specification (provided in Table 56 in this “Biodiesel” section), drain the tank, and flush the engine by running with high-quality fuel. Repeat the process until the system is clean. Refill with high-quality fuel following the recommendations provided in this “Fuel” section. For standby generator sets and emergency vehicles that use biodiesel, use fuel with oxidation stability 6 hours or more per “EN 14112” test method.

12. Biodiesel is prone to microbial contamination and growth due to the chemical nature. Microbial contamination and growth can cause corrosion in the fuel system and premature plugging of the fuel filter. Consult your supplier of fuel and additive for assistance in selecting appropriate anti-microbial additives. Use anti-microbial additives at the first sign of detection. The cleaning process will be more effective if the biocide is used before extensive growth of microorganisms.

13. Biodiesel can absorb and dissolve more water than diesel due to its chemical nature. Care must be taken to prevent water from contaminating the fuel and to remove water from fuel tanks. Water accelerates microbial contamination and growth and can cause system corrosion.

14. Biodiesel is not compatible with some metals. Biodiesel, in particular at blends of B20 and higher, will oxidize and form sediments upon long-term contact with lead, zinc, tin, copper, and copper alloys such as brass and bronze. These materials are typically avoided in the fuels systems. Consult your dealer for more information.

15. During the esterification and cleaning process of the biodiesel production, glycerol and unreacted oils (mono, di and tri glycerides) may remain in the biodiesel. If these undesirable components are above the allowed specification levels, they can cause issues such as filter plugging and injector deposits. It is critical that the biodiesel fuel is per the recommended specifications.

Note: The use of biodiesel at a B2 level improves the lubricity of the final blend by an estimated 66 percent. Increasing the blend level higher than B2 does not improve the lubricity any further.

Note: Use of biodiesel that is per or preferably exceeds the specifications can avoid the issues listed above and reduce the risks listed above.

Seasonal Operation

Seasonally operated engines should have the fuel systems, including fuel tanks, flushed with conventional diesel fuel before prolonged shutdown periods. Applications that should seasonally flush the fuel system include school buses (U.S.) and farm machinery.

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Perform the following process before shutting down the engine for prolonged periods:

1. Operate the engine until the fuel level in the tank is low.

2. Refill the fuel tank with high-quality conventional distillate diesel fuel.

3. Repeat steps 1 and 2 a minimum of two times before the engine is shut down for prolonged periods.

If distillate fuel is not available to operate the engine as described above, while accepting some risk, limit biodiesel to a maximum of B5. The use of biodiesel fuel of high oxidation stability can reduce the risks associated with prolonged storage. Follow the recommendations provided in this section and the guidelines given below to reduce the risk:

- Addition of a high performance diesel fuel oxidation stabilizer is recommended prior to engine shutdown for prolonged periods. The conditioner should be added when the engine is fueled. Preferably, the conditioner is added to the fuel as soon as possible after the fuel production.
- Addition of Cat Diesel Fuel System Cleaner, part number 343-6210, is recommended when the engine is first operated after the prolonged shutdown period and preferably for a total of two tanks of fuel. Follow the recommendations provided in the Cat Diesel Fuel System Cleaner section in the “Fuel” article in this Special Publication.
- For standby generator sets and emergency vehicles that use biodiesel, sample the fuel in the engine tank monthly. Test the fuel for acid number and oxidation stability. If the test results show that the fuel is degraded and not in specification (provided in Table 56 in this “Biodiesel” section), drain the tank, and flush the engine by running with high-quality fuel. **Repeat the process until the system is clean. Refill with high-quality fuel following the recommendations provided in this “Fuel” section. For standby generator sets and emergency vehicles that use biodiesel, use fuel with oxidation stability 6 hours or more per “EN 14112” test method.**

Consult your Caterpillar dealer for the availability of Cat Diesel Fuel System Cleaner. In case a fuel conditioner is needed to improve certain fuel properties, consult with your fuel supplier or with a reputable provider

Biodiesel Specification

Biodiesel fuel used for blending must meet the requirements in the following table, the requirements of ASTM “D5761” and/or “EN14214”.

The final blend of biodiesel as used in the engine must meet the requirements that are stated in Table 56 this “Biodiesel” article.

B100 intended for blending into diesel fuel that is expected to give satisfactory vehicle performance at fuel temperatures at or below −12° C (10.4° F) shall comply with a cold soak filterability limit of 200 seconds maximum. Passing “ASTM D6751” 200 seconds Cold Soak Filterability test limit does not guarantee cold performance for all biodiesel blends at all possible fuel temperatures, but biodiesel that fails this Cold Soak Filterability test requirement will produce biodiesel blends that will likely gel and plug fuel filters when fuel temperatures are below −12° C (10.4° F).

PROOF

Table 56

Cat Specification for Neat (B100) Biodiesel Blending Fuel				
Property	**Test Method, United States**	**Test Method, International**	**Units**	**Limits, B100 Blending Fuel**
Density at 15°C	“ASTM D1298”	“ISO 3675”	g/cm3	0.86-0.90
Viscosity at 40°C	“ASTM D445”	“ISO 3104”	mm2/s (cSt)	1.9-6.0
Flash Point	“ASTM D93”	“ISO 3679”	°C	93 minimum
Pour Point	-	-	-	6 °C (10 °F) minimum below ambient temperature
- Summer	“ASTM D97”	“ISO 3016”	°C	
- Winter				
Cloud Point	“ASTM D2500”		°C	Report
Sulfur Content(1)	“ASTM D5453”	“ISO 20846” “ISO 20884”	percent weight	0.0015(2) maximum
Distillation	-	-	-	-
T90	“ASTM D86”	“ISO 3924”	°C	360
Cetane Number	“ASTM D613”	“ISO 5165”	% evaporation	45 minimum
Sulfated Ash	“ASTM D874”	“ISO 3987” or "ISO 6245"	percent weight	0.02 maximum
Water and Sediment	“ASTM D2709”	“ISO 12937”	percent volume	0.050 maximum
Water	"ASTM D1796"	"EN ISO 12937"	% m/m	0.050 maximum
Copper Corrosion, 3 hours at 50°C	“ASTM D130”	“ISO 2160”	-	No. 1
Oxidation Stability	“EN 14112” or "EN 15751"	“EN 14112” or "EN 15751"	hours	3 minimum
Ramsbottom Carbon Residue on 10% bottoms	"ASTM D524"	“ISO 10370”	%-m/m	0.30 maximum
Carbon Residue, Conradson (CCR)	“ASTM D4530”	-	percent weight	0.05 maximum
Esterification	"ASTM D 7806" or "ASTM D 7371"	“EN 14103”	percent volume	97.5 minimum
Total Acid Number	“ASTM D664”	“EN 14104”	mg KOH/g	0.5 maximum
Methanol Content	“EN 14110”	“EN 14110”	percent weight	0.2 maximum
Monoglycerides	“ASTM D6584”	“EN 14105”	percent weight	0.8 maximum
Diglycerides	“ASTM D6584”	“EN 14105”	percent weight	0.2 maximum
Triglycerides	“ASTM D6584”	“EN 14105”	percent weight	0.2 maximum
Free Glycerin	“ASTM D6584”	“EN 14105”	percent weight	0.02 maximum
Total Glycerin	“ASTM D6584”	“EN 14105”	percent weight	0.240 maximum
Phosphorus Content	“ASTM D4951”	“EN 14107”	percent weight	0.001 maximum
Calcium plus Magnesium	“EN 14538”	“EN 14538”	ppm	5 maximum
Sodium plus Potassium	“EN 14538”	“EN 14538”	ppm	5 maximum
Cold Soak Filterability	“ASTM D7501”	–	seconds	360 maximum
Cleanliness	“ASTM D7619”	“ISO 4406”	–	(3)

(continued)

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(Table 56, contd)

Cat Specification for Neat (B100) Biodiesel Blending Fuel				
Property	**Test Method, United States**	**Test Method, International**	**Units**	**Limits, B100 Blending Fuel**
Total contamination	"EN 12662"	"EN 12662"	mg/kg	24 maximum
Appearance	"ASTM D4176"	-	-	Clear and Bright

(1) Methods for testing S are based on the sulfur content in diesel fuel used for blending. Consult with the fuel supplier and ensure the correct method has been used
(2) Sulfur level must be per local regulations. Higher S levels may be used where allowed.
(3) Recommended cleanliness level for fuel as dispensed into the engine fuel tank is ISO 18/16/13 or cleaner per the test methods listed. Refer to the contamination control Chapter for more details.

Note: Fuels that meet the most current revision level of “ASTM D6751” or “EN 14214” may be used for blending with an acceptable distillate fuel. The conditions, recommendations, and limits that are noted in this Biodiesel section apply.

Referenced Documents

Refer to the “Reference Material - Fuel” section of this Special Publication for the reference material for the information in this section.

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Cooling System Specifications

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General Coolant Information

SMCS Code: 1350; 1395

⚠ WARNING

The cooling system operates under pressure which is controlled by the radiator pressure cap. Removing the cap while the system is hot may allow the escape of hot coolant and steam, causing serious burns.

Before you remove the radiator cap, allow the system to cool. Use a thick cloth and turn the radiator cap slowly to the first stop to allow pressure to escape before fully removing the cap.

Avoid contact with coolant.

NOTICE
Every attempt is made to provide accurate, up-to-date information. By the use of this document, you agree that Caterpillar Inc. is not responsible for errors or omissions.

The information that is provided is the latest recommendations for the Cat diesel engines that are covered by this Special Publication. This information supersedes all previous recommendations which have been published for the Cat diesel engines that are covered by this Special Publication. Special fluids may be required for some engines. Continue to use these special products. Refer to the applicable Operation and Maintenance Manual.

This publication is a supplement to the Operation and Maintenance Manual. This publication does not replace the engine Operation and Maintenance Manual.

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

NOTICE
In order to avoid potential damage to your Cat machine and/or Cat engine, only purchase Cat fluids and Cat filters through your Cat dealer or Cat authorized outlets. For a list of authorized Cat parts outlets in your area, consult your Cat dealer.

If you purchase what appear to be Cat fluids and/or Cat filters through other outlets/sources, you are at a very high risk of purchasing counterfeit ("look-alike") products.

Counterfeit or "look-alike" products may visually appear the same as the original Cat product, but the product performance and internal quality will typically be very low.

Counterfeit or "look-alike" products have a very high likelihood of causing and/or allowing engine and/or machine compartment damage.

NOTICE
Commercial products that make generic claims of meeting "Cat" requirements without listing the specific Cat recommendations and requirements that are met may not provide acceptable performance. Commercial products may cause reduced engine and/or machine fluid compartment life. Refer to this Special Publication and refer to product specific Operation and Maintenance Manual for Cat fluids recommendations and requirements.

NOTICE
Never add coolant to an overheated engine. Engine damage could result. Allow the engine to cool first.

NOTICE
If the engine is to be stored in, or shipped to an area with below freezing temperatures, the cooling system must be either protected to the lowest outside temperature or drained completely in order to prevent damage caused by freezing coolant.

NOTICE
Never operate an engine without water temperature regulators in the cooling system. Water temperature regulators help to maintain the engine coolant at the proper operating temperature. Cooling system problems can develop without water temperature regulators.

If you operate the engine without the regulator, some coolant bypasses the radiator. This may cause overheating.

Note: Refer to the specific engine Operation and Maintenance Manual, "Maintenance Interval Schedule" for the correct interval for the replacement of the thermostat.

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Refer to Special Instruction, SEBD0518, “Know Your Cooling System” for more detailed information.

Many engine failures are related to the cooling system. The following problems are related to cooling system failures:

- Overheating
- Leakage of the water pump
- Plugged radiators or heat exchangers
- Pitting of the cylinder liners

These failures can be avoided with proper cooling system maintenance. Cooling system maintenance is as important as maintenance of the fuel system and the lubrication system. Quality of the coolant is as important as the quality of the fuel and the lubricating oil.

A coolant that is ready to use in the engine can also be referred to as “finished coolant”. A finished coolant is a coolant that has been diluted with appropriate amount of acceptable quality water.

Coolant is normally composed of the following elements:

- water
- additives
- glycol
- Embitterment: in coolants containing ethylene glycol to make the coolant taste bad.

Technical information for each of the coolant elements is provided in this Special Publication.

Water

NOTICE
Never use water alone as a coolant. Water alone is corrosive at engine operating temperatures. In addition, water alone does not provide adequate protection against boiling or freezing.

Note: In glycol-based coolants, Cat strongly recommends a minimum of 30 percent glycol in diesel engine cooling systems, with a minimum of 50 percent glycol recommended. Use only glycol-based coolants that meet one or more of the coolant specifications that are defined as preferred or acceptable in this Special Publication and that also comply with any additional requirements that are stated in this Special Publication (that is, chemical composition, the use of SCA, the use of Extender). Refer to the Operation and Maintenance Manual for your engine for any exceptions.

NOTICE
All Cat engines that are equipped with a Cat NOx Reduction System require a minimum of 50 percent glycol to help prevent cavitation damage and boiling of the engine coolant. These engines include Tier 4 engines.

Water in the water/glycol coolants is more effective than glycol alone in transferring heat.

Distilled water or deionized water is recommended to add to glycol or to water based coolants in engine cooling systems.

DO NOT use the following types of water in cooling systems: hard water, softened water that has been conditioned with salt and sea water.

If distilled water or deionized water is not available, use water that meets or exceeds the minimum acceptable water requirements that are listed in Table 57 .

Table 57

Cat Minimum Acceptable Water Requirements		
Property	**Maximum Limit**	**ASTM Test**
Chloride (Cl)	40 mg/L (2.4 grains/US gal)	“D4327”
Sulfate (SO_4)	100 mg/L (5.9 grains/US gal)	“D4327”
Total Hardness	170 mg/L (10 grains/US gal)	“D1126”
Total Solids	340 mg/L (20 grains/US gal)	“Federal Method 2540B”(1)
Acidity	pH of 5.5 to 9.0	“D1293”

(1) Total dissolved solids dried at 103° C (217° F) - 105° C (221° F), “Standard Method for the Examination of Water and Wastewater”, “American Public Health Association”, “www.apha.org”, “www.aphabookstore.org”, (888) 320-APHA.

For a water analysis, consult one of the following sources:

- Cat dealer
- Local water utility company
- Agricultural agent
- Independent laboratory

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Periodic analysis of water that is used to add to the coolant is recommended. Water quality can be affected by various factors including malfunctioning purification equipment, earthquakes, and droughts.

Additives

Additives help to protect the metal surfaces of the cooling system and can improve coolant performance. A lack of coolant additives, insufficient amounts of additives, or improper additives for the application can cause the following conditions to occur:

- Corrosion
- Formation of mineral deposits
- Rust
- Scale
- Pitting and erosion from cavitation of the cylinder liner
- Foaming of the coolant

Many additives are depleted during engine operation. These additives must be replaced periodically. Additives that can be added:

- ELC Extender to Cat ELC (Extended Life Coolant)
- ELC Extender to Cat ELI (Extended Life Inhibitor)
- Cat SCA (Supplemental Coolant Additive) to Cat DEAC (Diesel Engine Antifreeze/Coolant)
- · Cat SCA to Cat SCA in water finished coolant

Additives must be added at the proper concentration. Overconcentration of additives can cause the inhibitors to drop out-of-solution. The deposits can enable the following problems to occur:

- Formation of gel compounds
- Reduction of heat transfer
- Leakage of the water pump seal
- Plugging of radiators, coolers, and small passages

Glycol

Glycol in the coolant helps to provide protection against the following conditions:

- Boiling
- Freezing
- Water pump cavitation (ATAAC equipped engines)

For optimum performance, Cat recommends a 50 percent by volume of glycol in the finished coolant (also referred to as 1:1 mixture).

Note: Use a mixture that will provide protection against the lowest ambient temperature.

Most conventional heavy-duty coolant/antifreezes use ethylene glycol. Propylene glycol may also be used. In a 50 percent by volume of glycol in the finished coolant, ethylene and propylene glycol provide similar protection against freezing and boiling. Refer to tables 58 and 59 .

Table 58

Ethylene Glycol Concentration		
Concentration	**Freeze Protection**	**Boil Protection** (1)
20 Percent	−8° C (18° F)	102° C (216° F)
50 Percent	−37 °C (−34 °F)	106° C (223° F)
60 Percent	−52 °C (−62 °F)	111 °C (232 °F)

(1) Boiling protection is increased with the use of a pressurized radiator.

Do not use propylene glycol in concentrations that exceed 50 percent glycol because of the reduced heat transfer capability. Use ethylene glycol in conditions that require additional protection against boiling or freezing. Do not use ethylene glycol in concentrations that exceed 60 percent glycol.

Table 59

Propylene Glycol Concentration		
Concentration	**Freeze Protection**	**Boil Protection** (1)
50 Percent	−32 °C (−26 °F)	106 °C (223 °F)

(1) Boiling protection is increased with the use of a pressurized radiator.

Propylene glycol coolant that is used in the cooling systems for Cat diesel engines must meet ASTM D6210-06, “Fully-Formulated Glycol-Based Engine Coolant for Heavy-Duty Engines”. When propylene glycol coolant is used in heavy-duty diesel engines, a regular addition of SCA is required for protection against liner cavitation. Consult your Cat dealer for additional information.

Ethylene or propylene glycols used in cooling systems for Cat diesel engines must meet ASTM E1177-06, “Standard Specification for Engine Coolant Grade Glycol”.

Testing the Concentration of Glycol

To check the concentration of glycol, use the 245 - 5829 Coolant/Battery Tester/Refractometer or 360-0774 refractometer. The tester can be used with ethylene or propylene glycol.

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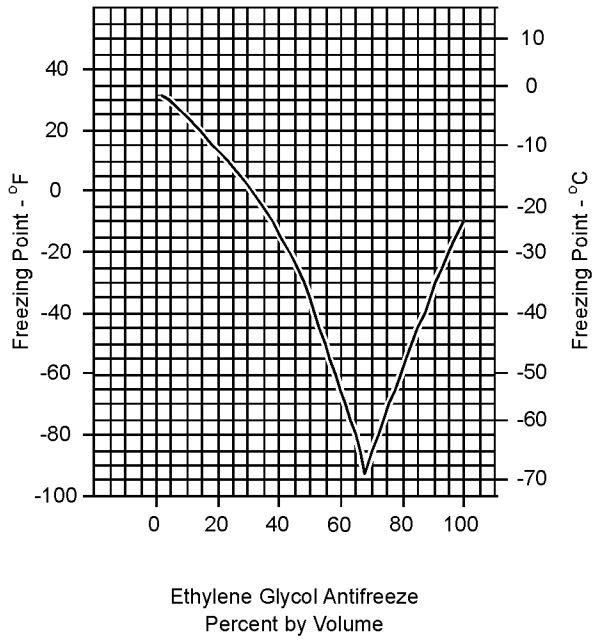


Illustration 3 g03387377

Approximate curve of the freezing point for a typical ethylene glycol solution.

Table 60

Freeze Protection for Antifreeze Concentrations(1)	
Protection to:	**Concentration**
−15 °C (5 °F)	30% glycol 70% water
−24 °C (−12 °F)	40% glycol 60% water
−37 °C (−34 °F)	50% glycol 50% water
−52 °C (−62 °F)	60% glycol 40% water

(1) Ethylene glycol-based antifreeze.

Alternative products that are used to protect from boiling or freezing of the engine coolant include:

- “1,3 propandiol” (PDO)
- glycerin
- mixtures of these alternative products with glycol

At the time of publication of this document, there currently exists no ASTM, “specifications” for coolants using these chemicals. **Until specifications are published and then evaluated by Cat, use of PDO and glycerin or glycerin/glycol coolants are not recommended in Cat cooling systems.**

Embitterment

Ethylene glycol is a toxic chemical with a naturally sweet taste. In order to avoid accidental excessive ingestion by humans or animals, coolants may contain embittering agents that make the coolant taste bad. **All Cat glycol containing coolants (Cat ELC, Cat DEAC, and Cat NGEC) are embittered.** Embittering agents have no beneficial or detrimental effect on coolant performance or characteristics.

Coolant Terminology

Extended Life Coolant (ELC) – A coolant that relies largely on organic inhibitors for corrosion and cavitation protection. Carboxylate is an example of organic corrosion and cavitation inhibitors. Cat ELC and Cat ELI in water are extended life coolants that also include nitrites and molybdates for increased cavitation protection.

- Commercial extended life coolants containing silicate do not meet the additional requirements set in this Special Publication for coolants claiming to meet Cat EC-1 specification.
- Do not use commercial extended life coolants with more than 125 ppm silicon (present in the coolant in the form of silicate)
- Extended life coolants that meet “ASTM D6210-06” may be used at the recommended maximum coolant service life intervals stated in this Special Publication for coolants that meet the ASTM specifications.

Conventional coolant – A coolant that relies largely on inorganic inhibitors for corrosion and cavitation protection. Silicates and nitrites are examples of inorganic inhibitors. Conventional coolants are also referred to as heavy-duty coolants, heavy-duty fully formulated coolants, or traditional coolants. In order to be used in most Cat cooling systems, conventional coolants must meet “ASTM D6210-06”. Cat DEAC and Cat SCA in water (also referred to as SCA/Water coolant) are conventional coolants

Supplemental Coolant Additive (SCA) – SCA is a general term for a concentrated inorganic inhibitor package. SCA is used for three different purposes:

- to precharge a new conventional coolant that is not fully Formulated. Cat DEAC is fully formulated and does not require SCA

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- to provide corrosion protection in water/SCA cooling systems
- to recharge an in service conventional coolant on a maintenance basis to maintain proper inhibitor levels

Extender – An inhibitor package that is added to extended life coolants, to recharge an in-service coolant. Extenders, typically, only should be added at one half the service life of the coolant.

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Coolant Recommendations

SMCS Code: 1350; 1352; 1395

The finished coolants that are recommended or acceptable for use in Cat diesel engines are given in Table 61 below:

Table 61

Recommendations for Finished Coolants for use in Cat engines				
Coolant Type	**Recommendations**	**Product**	**Service hours (1)(2)**	**Required Maintenance(3)**
Cat ELC, Cat ELI, or commercial coolant that meets Cat EC-1	Preferred	Cat ELC	12000 hours or 6 years	Add Cat ELC Extender at 6000 service hours or one half of service life
		Cat ELI (water based) (4)	12000 hours or 6 years	Add Cat ELC Extender at 6000 service hours or one half of service life
	Min requirements	Cat EC-1 specification and “ASTM D6210” and Organic Additive Technology (OAT) based on a combination of a monocarboxylic acid and a dicarboxylic acid. Free of phosphate, borate and silicate. Tolyltriazole: minimum typical concentration of 900 ppm Nitrite: minimum typical concentration of 500 ppm in new coolants	6000 hours or 6 years	Add Extender at 3000 service hours or one half of service life
Cat DEAC, Cat SCA, conventional coolants, and commercial extended life coolants that do not meet EC-1	Acceptable	Cat DEAC	3000 hours or 3 years	SCA at maintenance intervals
		Cat SCA (water based) (4)	3000 hours or 2 years	SCA at maintenance intervals

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(Table 61, contd)

Recommendations for Finished Coolants for use in Cat engines				
	Min requirements for fully formulated Heavy Duty Commercial coolants	"ASTM D6210" and Nitrite (as NO2) concentration: Minimum of 1200 ppm (70 grains/US gal) and maximum of 2400 ppm (140 grains/US gal) Silicon concentration: minimum of 100 ppm and maximum of 275 ppm	3000 hours or 2 years	SCA at maintenance intervals
	Min requirements for Commercial coolants requiring SCA precharge.	ASTM D4985 and Nitrate (as NO2) concentration: Minimum of 1200 ppm (70 grains/US gal) and maximum of 2400 ppm (140 grains/US gal) Silicon concentration: minimum of 100 ppm and maximum of 275 ppm	3000 hours or 1 year	SCA at initial fill and SCA at maintenance intervals
	Min requirements for SCA and water (5)	Commercial supplemental coolant additive and water having Nitrite (as NO2) concentration: Minimum of 1200 ppm (70 grains/US gal) and maximum of 2400 ppm (140 grains/US gal) Silicon concentration: minimum of 100 ppm and maximum of 275 ppm	3000 hours or 1 year	Per manufacturer recommendations

(1) New Coolants at 50 volume percent diluted. Coolants that are prediluted at the coolant manufacturer must be diluted with water that meets Reagent 4 "ASTM D1193" requirements.
(2) Maintain the in-service coolant at the given limits.
(3) For appropriate maintenance procedures, refer to the details given in this Chapter. For applications that allow the use of Cat ELI in water, a minimum of 7.5 percent of Cat ELI is recommended. For applications that allow the use of SCA and water, a minimum of 6 percent to a maximum of 8 percent concentration of Cat SCA are recommended.
(4) Water-based coolants are not allowed for use in machines that has NOx reduction aftertreatment devices, in engines that has AATAC and in Marine engines that have SWAC
(5) There are currently no industry standards to define the quality of water-based conventional coolants. To control the quality of SCA and water coolants, the commercial SCA additive package should pass ASTM D6210 when this package is used in a glycol based coolant. Do not use a commercial SCA additive package that only meets the ASTM D3306 or equivalent specification when used in a glycol based coolant.

Table 62

Special Requirements(1)	
Cat diesel engines equipped with air-to-air aftercooling (ATAAC)	Minimum of 30% glycol is required. 50% Glycol is recommended. Water alone or water with SCA or with ELI is NOT allowed.
Cat C7-C32 marine engines with engine mounted heat exchangers and raw water aftercoolers (SWAC(2))	
Cat 3618 engine model	Water based ELI is recommended. Water based SCA is acceptable. Glycol-based coolants are NOT allowed.
Cat C18 SCAC(3), Cat C32 SCAC, and Cat 3500C Tier 3 and Cat 3500E Tier 4(4) certified marine engines	Maximum of 20% glycol is allowed in the aftercooler circuit.(5) (not applicable to the jacket water circuit, which is capable of 50% glycol mixture)

(1) Water based ELI and SCA are not recommended for use in applications that require freeze protection.
(2) SWAC: seawater aftercoolers
(3) SCAC: Separate Circuit Aftercooler
(4) 43° C SCAC inlet temperature is allowed when using treated water only.
(5) For freezing protection of −5° C (23° F) temperatures and lower, consult your Cat dealer for guidance.

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